Winter Internship

On

"Statistical Study of Heart Attacks"

Submitted to

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In partial fulfillment of requirements for the award of the Degree of

(M.Sc. Data Science)

By

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LIST OF FIGURES

LIST OF TABLES

TABLE OF CONTENTS

1. INTRODUCTION

Heart attacks, are one of the most prevalent and prominent health concerns for individuals around the world. A recent report by World Health Organization (WHO) claims that, “cardiovascular diseases are primary cause of death all over the world. This death results from the significant interruption of blood flow to the heart, leading to severe damage if quick action is not taken to restore perfusion.

The rising incidence of heart attacks can be attributed to multi-faceted complex factors. These factors can be physiological like hypertension, hypercholesterolemia, diabetes mellitus; unhealthy habits like smoking, poor nutrition, and sedentary lifestyle; stress, irregular health screenings, infrequent exercise; and alcohol intake. Furthermore, older patients, males, and having a family history are also very important counselling factors in estimating the risk. You cannot grasp the hundreds of these factors without the use of statistics.

Through the use of a variety of statistical tools which include regression analysis, correlation studies, chi-square tests, and hypothesis testing we will look at the role of both changeable and unchangeable risk factors in heart attack incidence. From this data we aim to put forth which is the most relevant information for early detection and very effective prevention in clinical and public health settings. By looking at trends in the data we aim to identify which are the primary players in heart attack incidence and how different variables play together. This research looks at the issue of the epidemiology of heart attacks as well as we put forward into the field targeted interventions and policies to determine what we can do to reduce the burden of cardiovascular disease in the population. Also this statistical study is a step toward that which improves awareness, diagnosis and prevention of heart attacks which in turn we hope will be a foundation for future research and health care strategies aimed at saving lives and improving quality of life.

2. REVIEW OF LITERATURE

Heart attacks, in specifically, are among the most common causes of death around the world due to cardiovascular disorders (CVDs). Around 17.9 million people die each year from CVDs, with heart attacks accounting for a significant number of these deaths, estimates the World Health Organization's estimates (2021). As a result, an enormous amount of research has been done to use machine learning and statistical methods to understand and project the risk of heart attacks in the future.

The Framingham Heart Study, one of the groundbreaking investigations in this field, found critical risk factors like age, blood pressure, cholesterol, smoking, diabetes, and family history (Mahmood et al., 2014). Since that time predictive models have began employing these features as an standard inputs.
As machine learning has grown, ensemble techniques such as Random Forest have demonstrated better predictive power., Rathee et al. (2023) discovered that Random Forest outperformed Logistic Regression (85.25%) when it came to of precision (90.16%).

Also, Miah et al. (2023) evaluated some models. They determined that XGBoost had an accuracy of 92.7%, while Random Forest and Logistic Regression maintained to perform in competition. When looking into the connections within heart attack risk and variables like blood pressure, BMI, cholesterol, and smoking status, methods for visualization like the distribution charts, box plots, and correlation heatmaps that are commonly used.
In summary, The research indicates that machine learning models and statistical techniques are both useful for predicting heart attacks. Although Random Forest frequently provides greater predictive power, logistic regression is frequently trusted in clinical settings due to its transparency. Drawing from these studies, this project compares the two methods in order to determine which model, based on patient health data, is most effective at predicting the risk of a heart attack.

3. DATA PREPARATION

3.1 Dataset Description

This heart attack prediction info comes from 8,763 people and has 26 pieces of info for each person, all to help guess who might have a heart attack. Each person has their own ID.The info covers things like: how old they are, if they're male or female, what country they're from, and which part of the world they live in.

There's also medical stuff like cholesterol, blood pressure, heart rate, if they have diabetes, if heart problems run in their family, if they're on meds, their BMI, and triglyceride levels. The info also looks at habits like smoking, if they're overweight, drinking, how much they work out each week, what they eat, how stressed they are, how much they sit around all day, how many days a week they're active, and how much they sleep.

Income is also included as a socio-economic indicator. The Blood Pressure field is stored as a string in systolic/diastolic format (e.g., “130/80”) and may require preprocessing.

The Heart Attack Risk is either high (1) or low/none (0). Most info is in number form, but some categories exist like Sex, Diet, and Continent.

The data is pretty clean; nothing's missing. It covers different people from many countries and continents. This lets us really dig in and build models to guess heart attack risks based on what affects them.

3.2 Data Collection

The data was collected from Kaggle, it is an online community platform for information researchers and machine learning devotees to discover and distribute dataset.

Table 1: The original dataset

	Patient ID	Age	Sex	Cholesterol	Blood Pressure	Heart Rate	Diabetes	Family History	Smoking
0	BMW7812	67	Male	208	158/88	72	0	0	1
1	CZE1114	21	Male	389	165/93	98	1	1	1
2	BNI9906	21	Female	324	174/99	72	1	0	0
3	JLN3497	84	Male	383	163/100	73	1	1	1
4	GFO8847	66	Male	318	91/88	93	1	1	1

5 rows × 26 columns

4. EDA (EXPLORATORY DATA ANALYSIS)

The handle of classifying the information utilizing statistical and visual methods in arrange to highlight noteworthy highlights for extra examination is known as exploratory data analysis, or EDA. This incorporates examining the dataset from different viewpoints and giving a portrayal and rundown of its substance without accepting anything.

Look at the Information: Gather data around the information, like number of columns and also the type of data each columns contains. This incorporates understanding single factors and their distributions.

Clean the Information: Settle issues like lost or inaccurate values. Preprocessing is important for ensuring that the information is prepared for investigation and modelling.

Make Rundowns: Summarize the information to get a common thought of its substance, such as normal values, common values, or esteem dispersions. Calculating quantiles and checking for skewness can give experiences into the data's distribution.

Visualize the Information: Utilize intuitively charts and charts to spot patterns, designs, or irregularities. Bar plots, diffuse plots, and other visualizations help get it variables' connections. Python libraries are commonly used.
Ask Questions: Define questions based on your perceptions, such as why certain information focuses vary or if there are connections between diverse parts of the data.

Find Answers: Burrow more profound into the information to reply these questions, which may include assist examination or making models, counting relapse or straight relapse models.

4.1 Preprocessing Data

The following steps were carried out to ensure the data was clean, consistent, and ready for machine learning algorithms

4.2 Handling Missing Values

The dataset was checked for missing or null values using python code,
df.isnull().sum()
The dataset contains no missing values.

4.3 Outliers Removal

- **Original data shape:** `(8763, 26)`
- **After outlier removal:** `(8763, 26)`

There were no extreme outliers based on the standard IQR criteria in those features.

4.4 **Descriptive Statistics**

	Age	Cholesterol	Heart Rate	Diabetes	Family History	Smoking	Obesity	Alcohol Consumption	Exercise Hours Per Week	Previous Heart Problems	Medication Use	Stress Level	Sedentary Hours Per Day	Income
count	8763.000000	8763.000000	8763.000000	8763.000000	8763.000000	8763.000000	8763.000000	8763.000000	8763.000000	8763.000000	8763.000000	8763.000000	8763.000000	8763.000000
mean	53.707977	259.877211	75.021682	0.652288	0.492982	0.896839	0.501426	0.598083	10.014284	0.495835	0.498345	5.469702	5.993690	158263.181901
std	21.249509	80.863276	20.550948	0.476271	0.499979	0.304186	0.500026	0.490313	5.783745	0.500011	0.500026	2.859622	3.466359	80575.190806
min	18.000000	120.000000	40.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.002442	0.000000	0.000000	1.000000	0.001263	20062.000000
25%	35.000000	192.000000	57.000000	0.000000	0.000000	1.000000	0.000000	0.000000	4.981579	0.000000	0.000000	3.000000	2.998794	88310.000000
50%	54.000000	259.000000	75.000000	1.000000	0.000000	1.000000	1.000000	1.000000	10.069559	0.000000	0.000000	5.000000	5.933622	157866.000000
75%	72.000000	330.000000	93.000000	1.000000	1.000000	1.000000	1.000000	1.000000	15.050018	1.000000	1.000000	8.000000	9.019124	227749.000000
max	90.000000	400.000000	110.000000	1.000000	1.000000	1.000000	1.000000	1.000000	19.998709	1.000000	1.000000	10.000000	11.999313	299954.000000

Table 2: Descriptive statistics

The dataset compiles 8,763 people's health-related characteristics. The average age is approximately 54 years, and the average heart rate and cholesterol are 75 bpm and 260 mg/dL, respectively. Ninety percent smoke, fifty percent are obese, and sixty-five percent have diabetes, all of which are health risk factors.The wide range of income, from ₹20,062 to ₹299,954, suggests that participants come from a variety of socioeconomic backgrounds.

4.5 **Univariate Analysis**

Creating Bar Charts with Categorical Columns We employed bar plots made with the Seaborn library to display these categorical columns. To iterate through each categorical column and plot the distribution of its values, a loop was put in place.

4.5.1 Sex

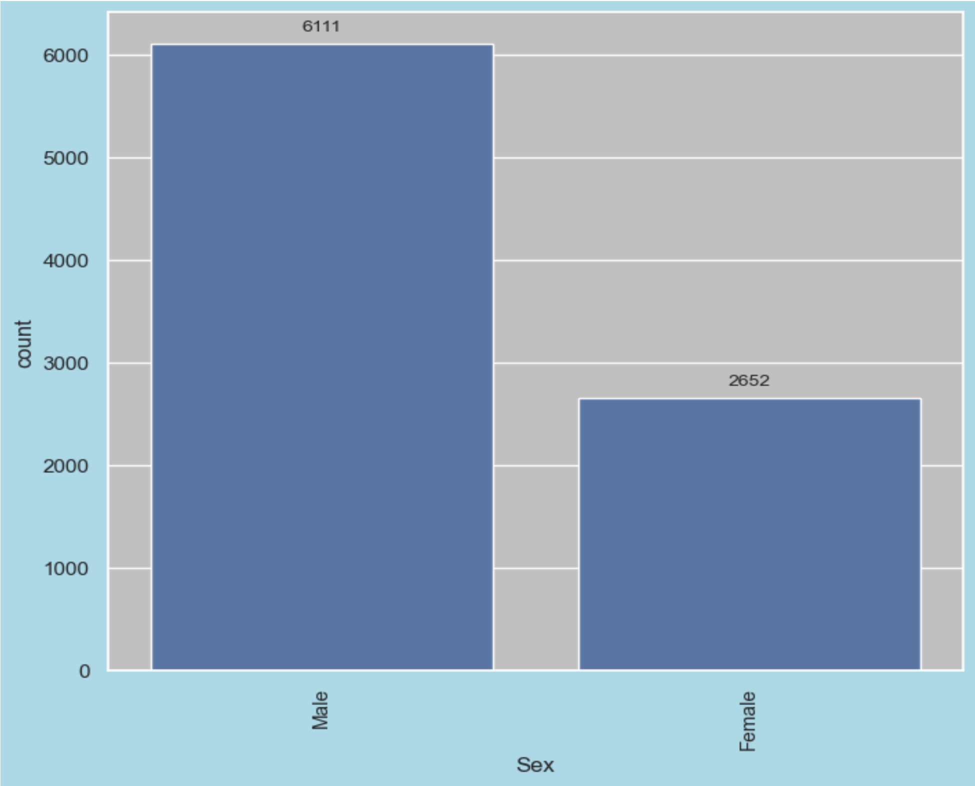


Figure 1: Distribution of the 'sex' variable in the dataset

The distribution of the 'Sex' variable in the dataset is shown in the bar plot. A summary of the interpretation is provided below:

- There are 6111 male patients, a much higher number than female patients.
- Compared to males, female patients are underrepresented in the dataset, as evidenced by their 2652 count.

Thus, we can say that,

- The distribution of the genders is obviously unbalanced.
- If not handled appropriately, this could potentially skew the analysis or predictive models (e.g., through sampling techniques or stratification).
- It is crucial to investigate if this imbalance results from bias in data collection or from real-world sampling (for example, more male patients coming in for heart exams).

4.5.2 Dietary habits

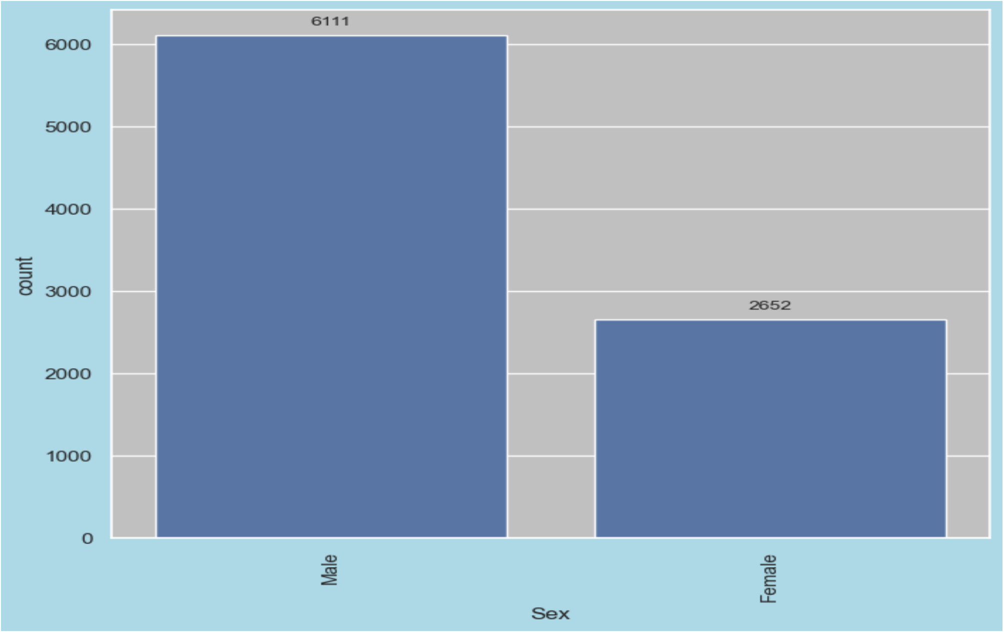


Figure 2:Distribution of the 'dietary habits' variable in the dataset

The distribution of eating patterns among the dataset's participants, divided into three groups, is displayed in this bar plot:

- A nutritious diet: Count = 2960

The fact that there are the most members of this group suggests that the majority of the dataset's participants eat a healthy diet.

- Average Diet = 2912

This indicates that a sizeable portion eats a moderately balanced diet because their numbers are very close to those of the healthy group.

- Diet Unhealthy: Count = 2891

Although there are a few fewer people in this group, the total is still quite large. Thus, we can say that, With very slight variations in counts, the distribution is fairly balanced across all three categories.

3.5.3 Distribution by continents

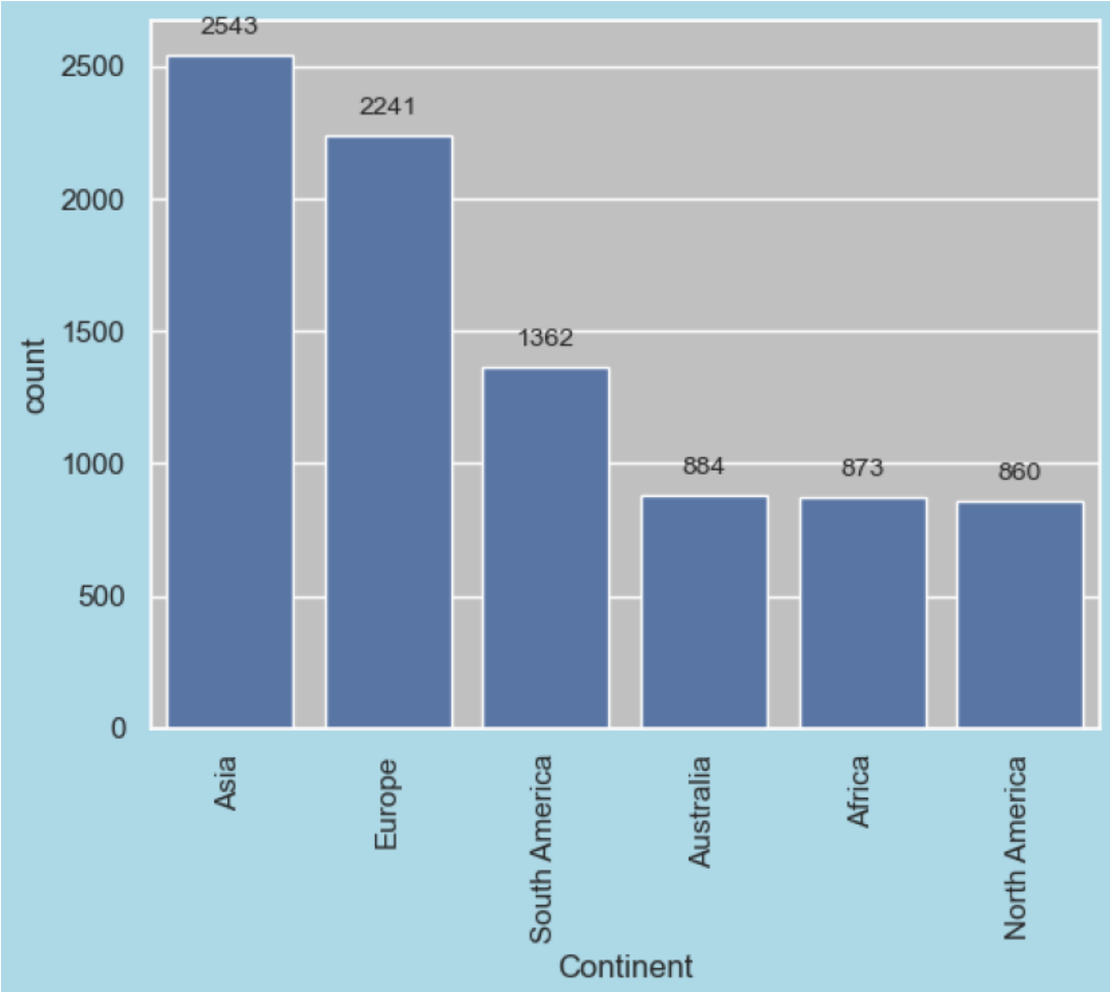


Figure 3: Distribution of continents

The number of people in the dataset grouped by continent is shown in the bar plot. With 2543 people, Asia is the region with the largest representation.

- Europe: 2241, second-highest.
- South America: 1362 people, a moderate amount of representation.

Australia, Africa, and North America are similarly underrepresented, with 884, 873, and 860 people, respectively.

Therefore, we can conclude that, The distribution is noticeably out of balance across continents.The results may not be as applicable across continents due to the data's significant bias toward Asia and Europe. Insights specific to North America, Africa, and Australia may be limited due to their underrepresentation.

3.5.4 Distribution of individuals across different countries

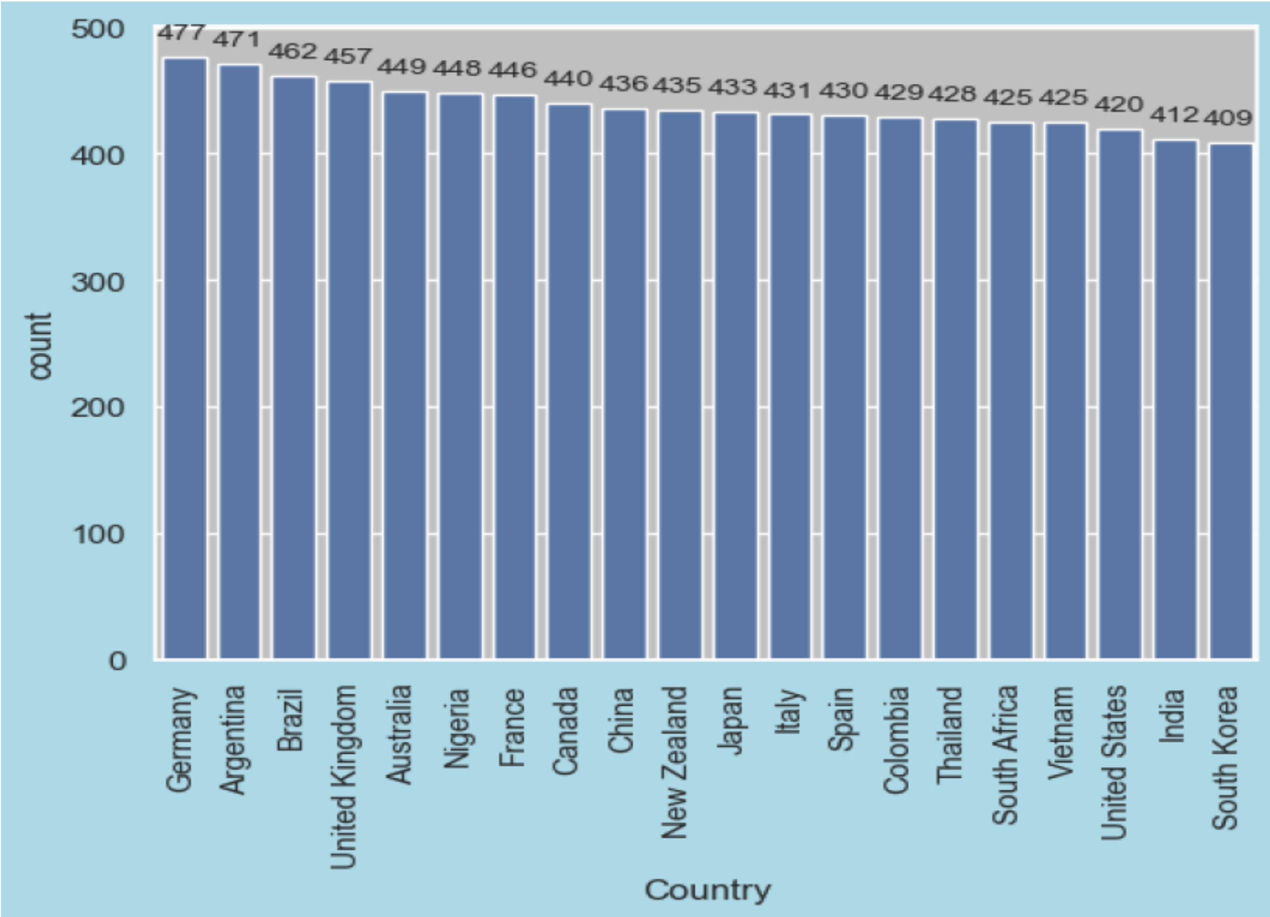


Figure 4: Distribution of countries

The top three most represented nations are:

- 477 people from Germany
- Argentina: 471 people
- 462 people from Brazil

The three least represented nations (though they are still fairly close to others) are:

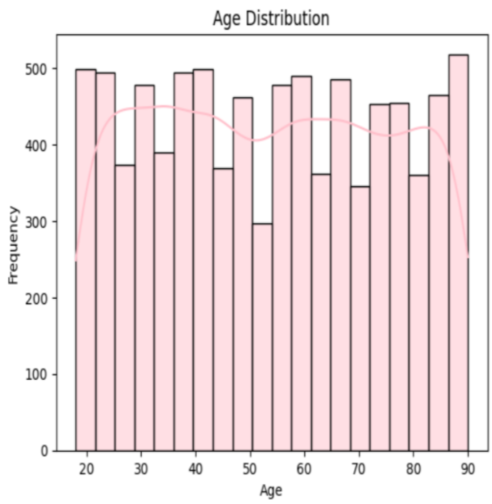
- India: 412 people
- United states : 420 people.
- 409 people from South Korea

Thus, we can deduce that,

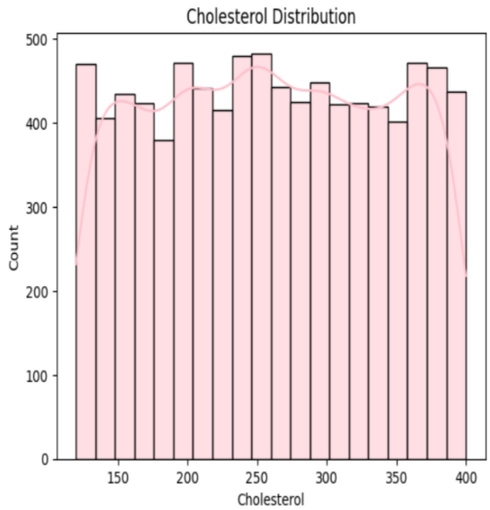
- The 20 nations are evenly distributes among the dataset.
- No country is overly dominant or underrepresented, despite minor differences.

This balance can help guarantee more globally applicable and generalizable results, particularly when analyzing trends by nation.

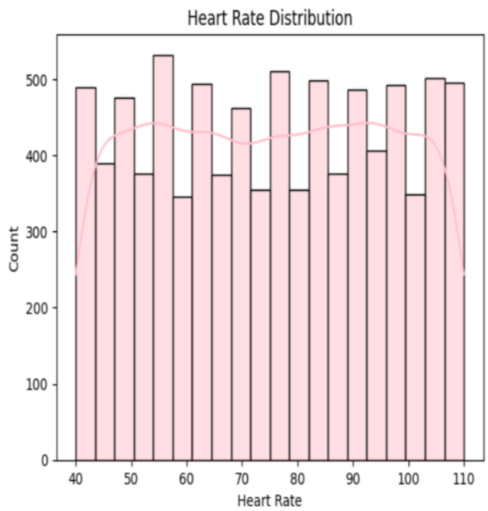
- **SOME MORE DISTRIBUTION PLOTS**.



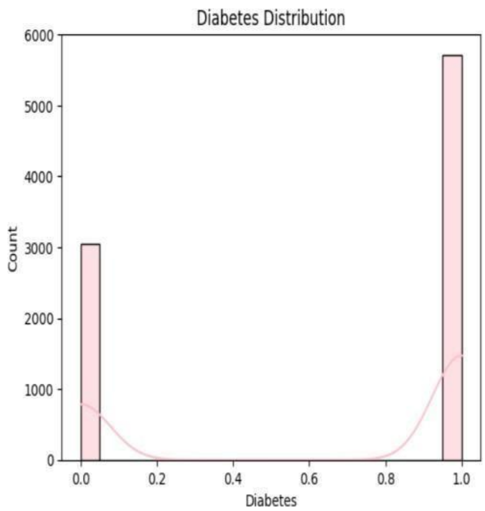
(a)



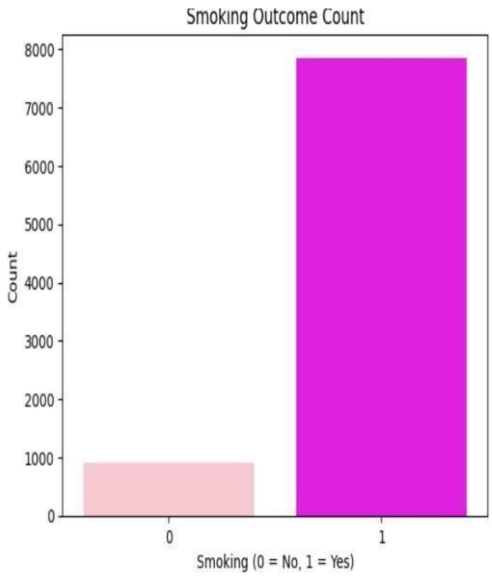
(b)



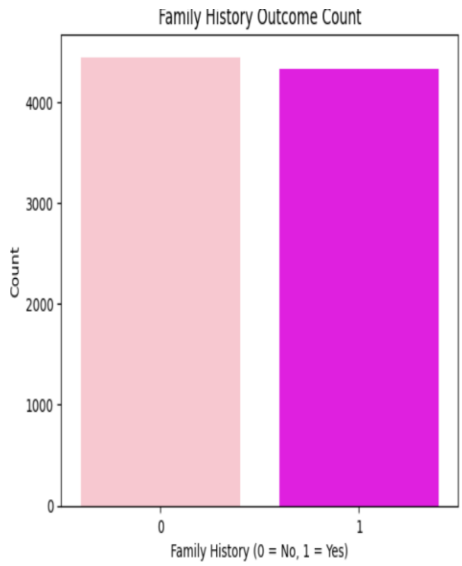
(c)



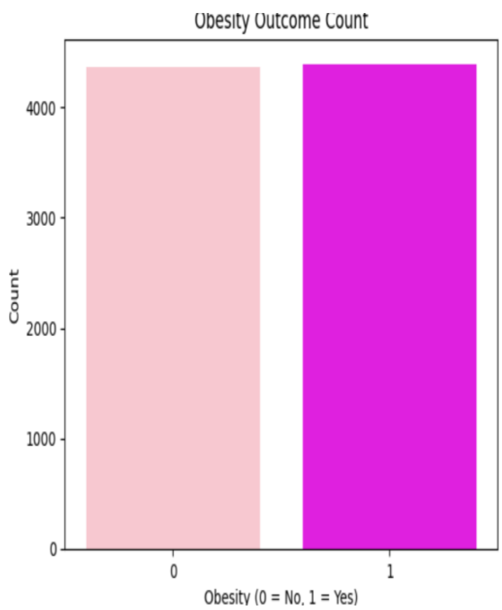
(d)



(e)



(f)



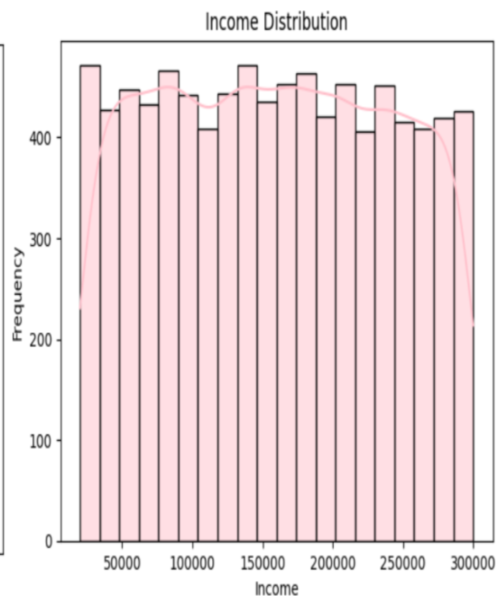


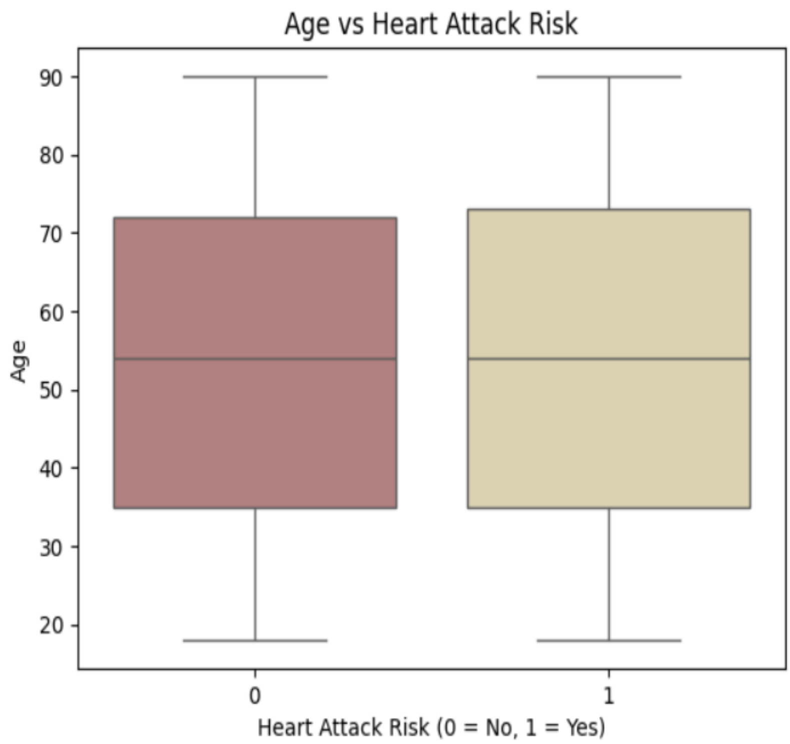
Figure 5: fig (a) to (h) shows the count for different variables

(g) (h)

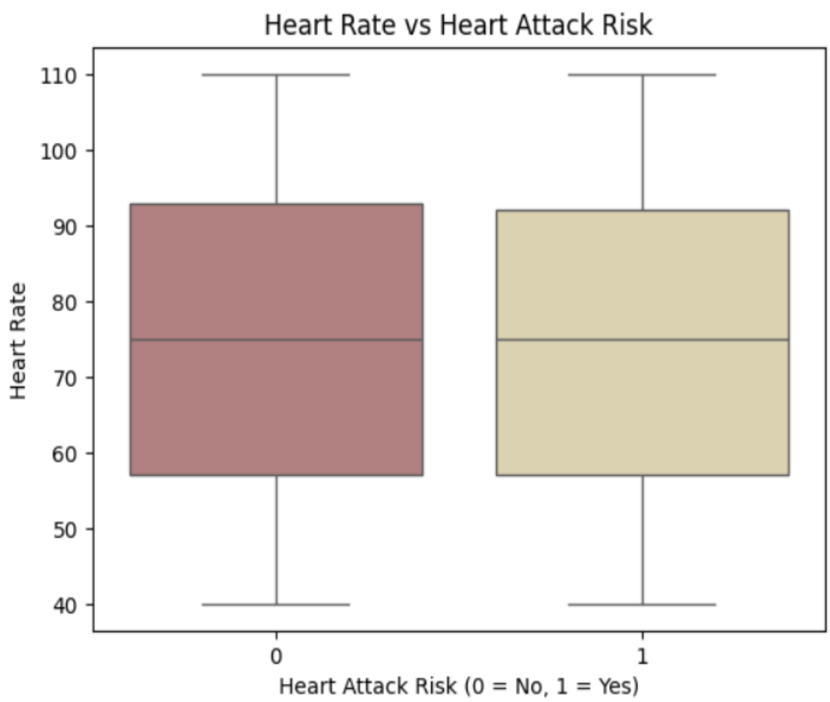
4.6 Bivariate Analysis

Two variables are compared in this section, particularly with regard to the target variable (heart attack risk).

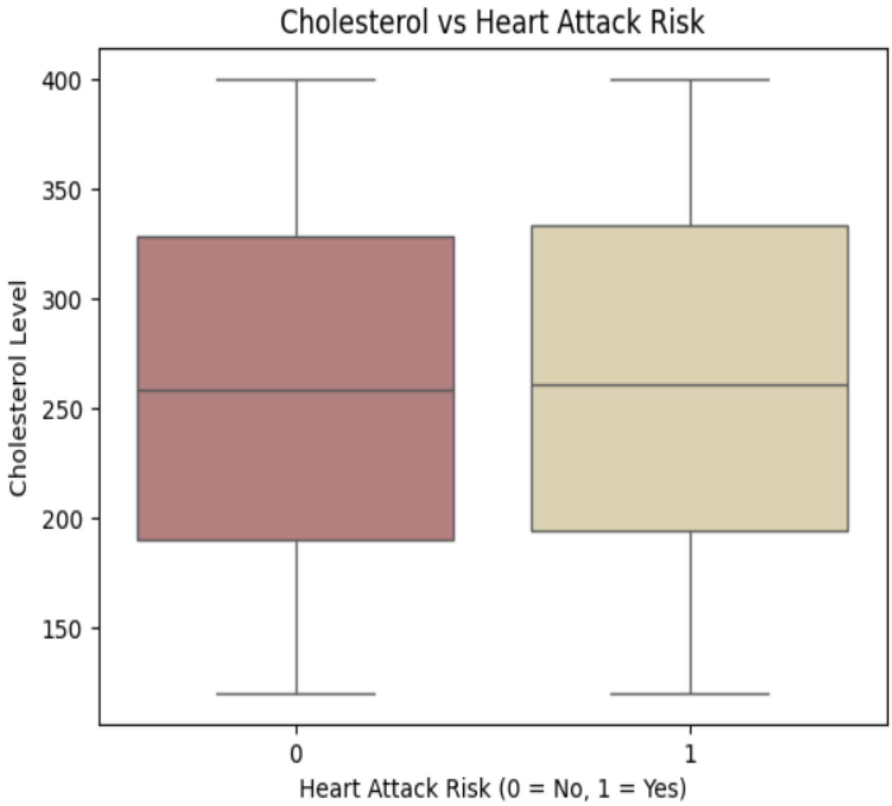
4.6.1 Box Plots



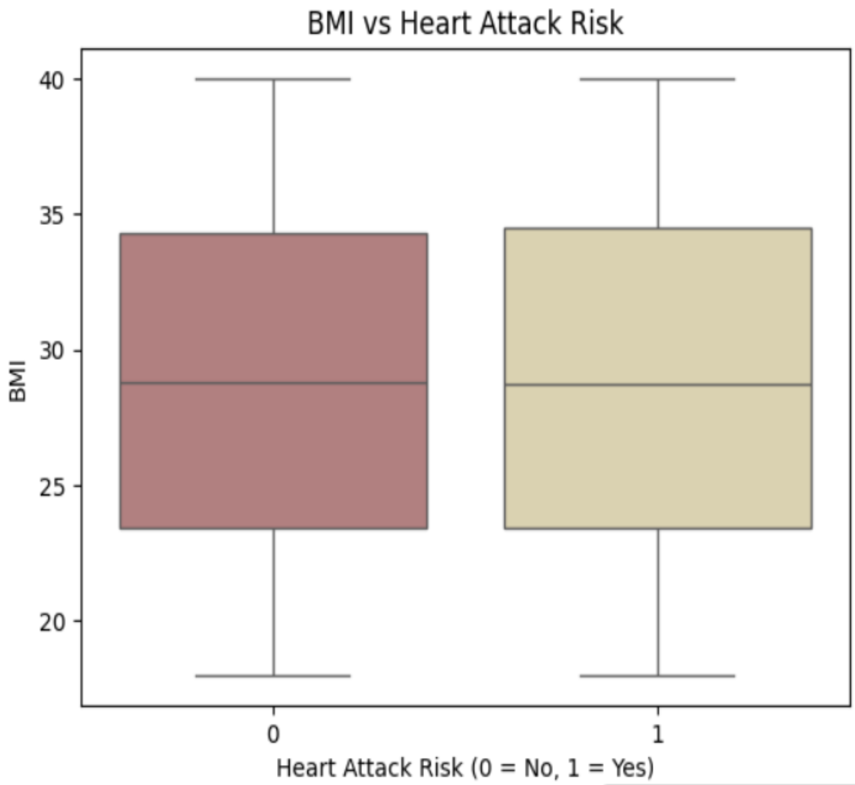
(a)



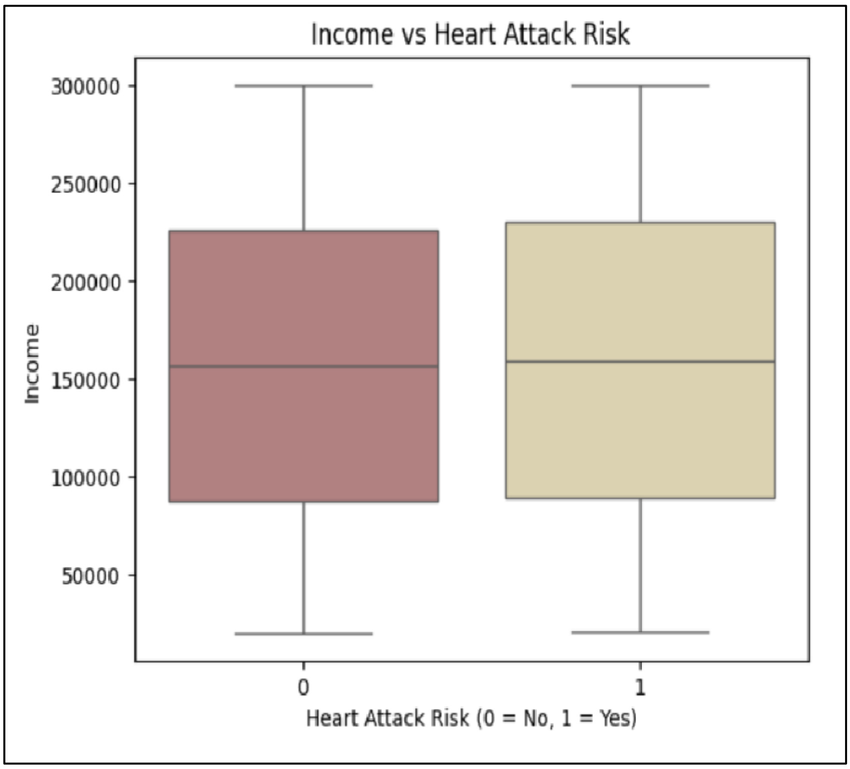
(b)


Cholesterol vs Heart Attack Risk
400
350
300
250
200
150
Cholesterol Level
0
1
Heart Attack Risk (0 = No, 1 = Yes)

(c)


BMI vs Heart Attack Risk
40
35
30
25
20
BMI
0
1
Heart Attack Risk (0 = No, 1 = Yes)

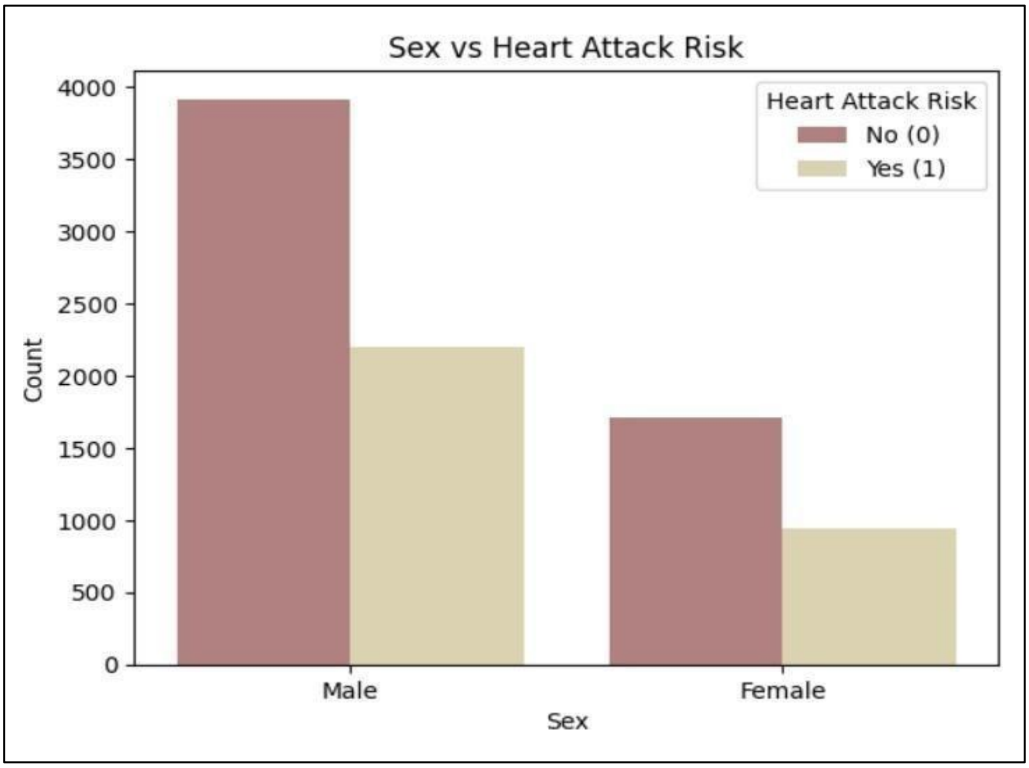
(d)



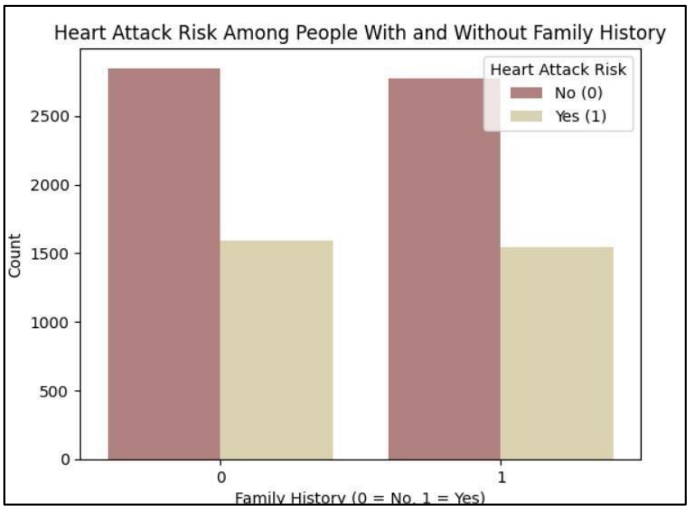
(e)

Figure 6: shows the box plots for different variables

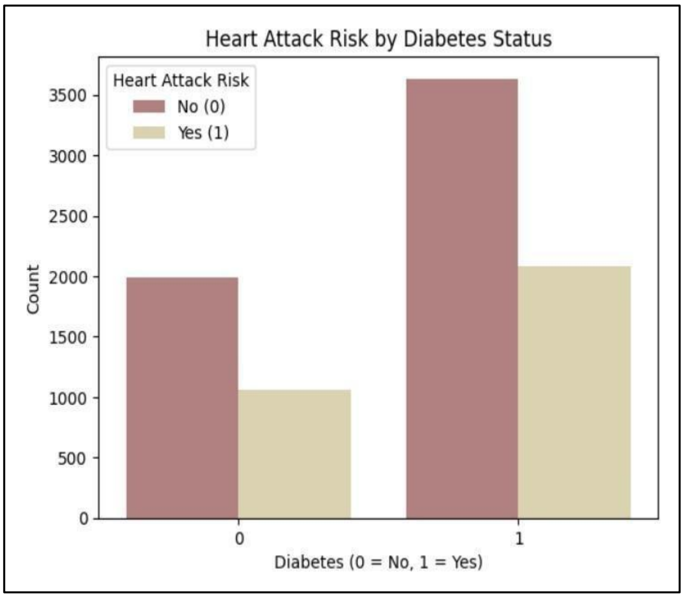
4.6.2 Count plot for Categorical Features



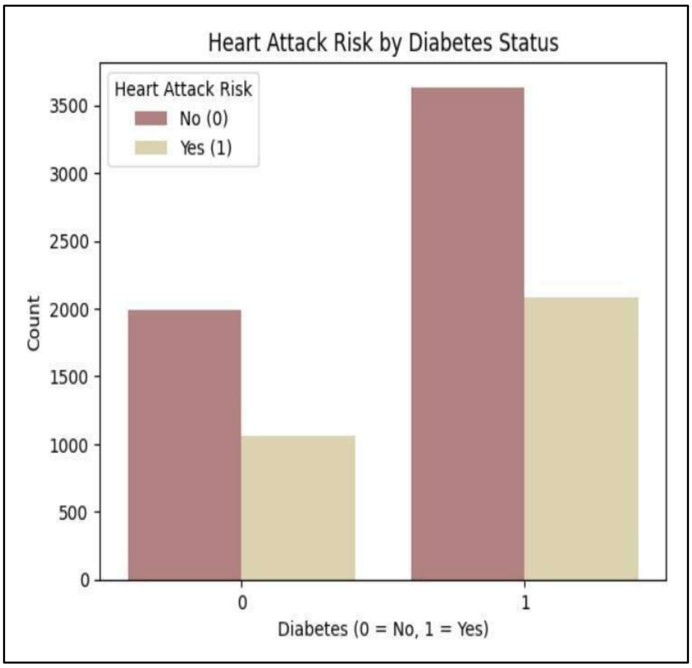
(a)

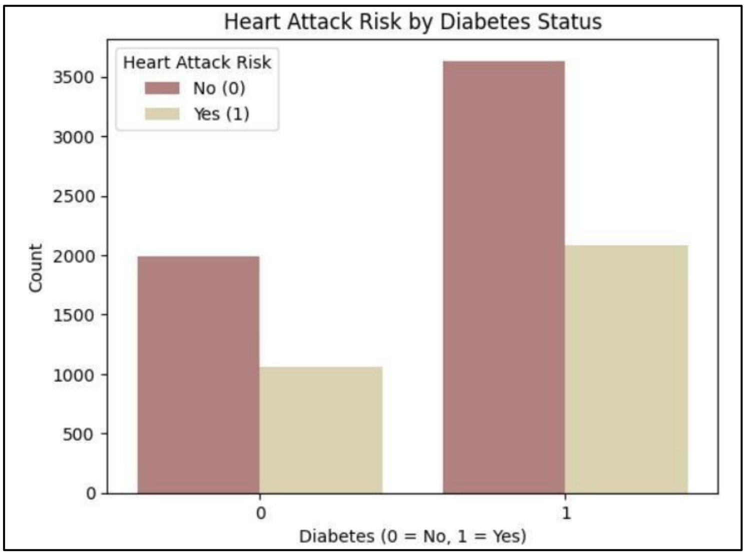

Heart Attack Risk Among People With and Without Family History
Heart Attack Risk
No (0)
Yes (1)
Count
0
500
1000
1500
2000
2500
0
1
Family History (0 = No, 1 = Yes)

(b)

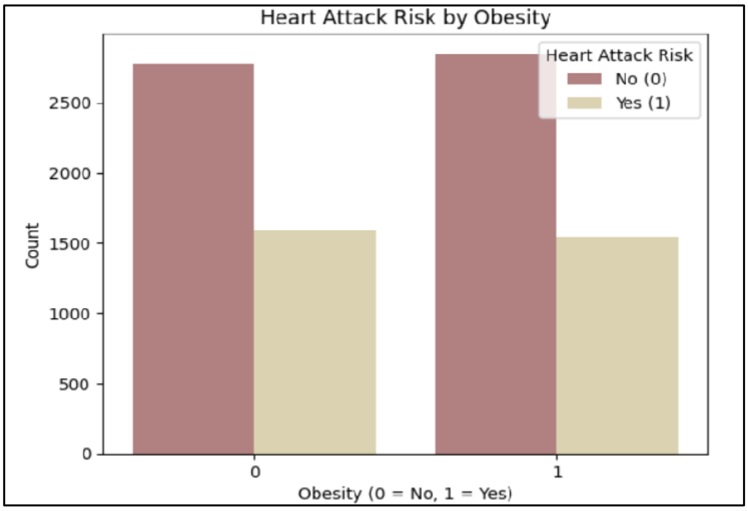

Heart Attack Risk by Diabetes Status
Heart Attack Risk
No (0)
Yes (1)
Count
0
500
1000
1500
2000
2500
3000
3500
0
1
Diabetes (0 = No, 1 = Yes)

(c)

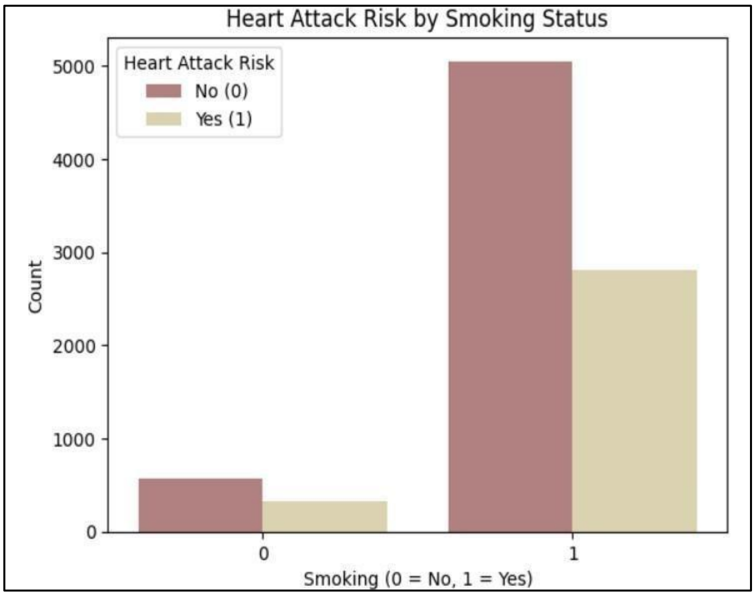

Heart Attack Risk by Diabetes Status
Heart Attack Risk
No (0)
Yes (1)
Count
0
500
1000
1500
2000
2500
3000
3500
0
1
Diabetes (0 = No, 1 = Yes)



(e)



(f)



(g)

Figure 7: shows the count plots for different variables with heart attacks

5. IDENTIFYING THE RESPONSE VARIABLE

Heart Attack Risk, which measures a person's chance of having a heart attack, is the response variable in this study. Age, cholesterol, heart rate, BMI, exercise routines, sedentary hours, smoking, alcohol use, and so forth are some of the predictor variables.

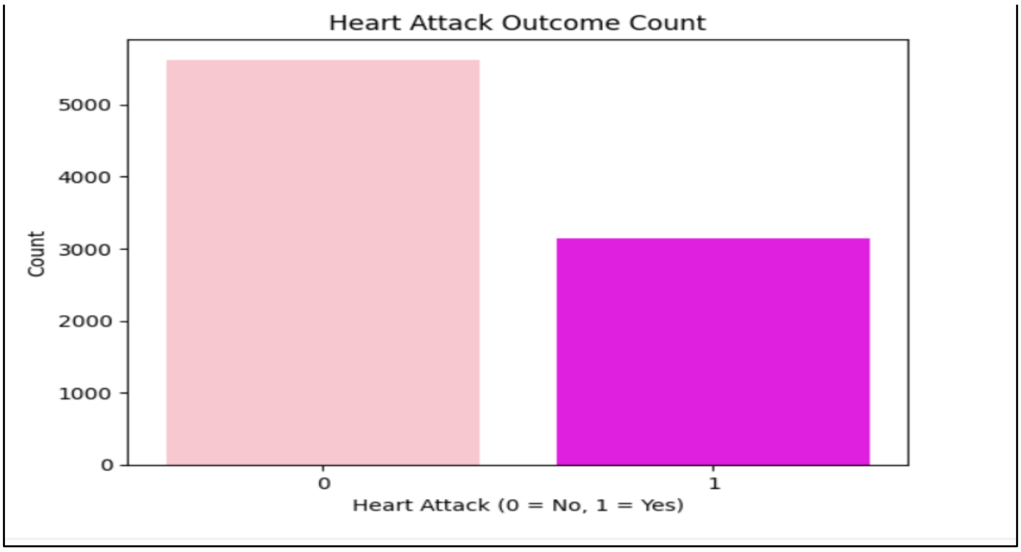


Figure 8 : Shows the distribution of heart attack risk among individuals

6. STUDYING FACTORS

In order to find the relationship between the target variables a correlation matrix was created (heart attack risk).Values in the matrix range from -1 (perfect negative correlation) to +1 (perfect positive correlation) indicating how strongly each variable is linearly related to the others.

From the matrix we can say:

- The majority of variables had weak relationships with the risk of a heart attack.
- There was a very weak positive correlation between heart attack risk and cholesterol, heart rate, and triglycerides.
- The risk of having a heart attack was not significantly correlated with BMI, weekly exercise hours, daily sedentary hours, or smoking.
- Age and smoking showed the strongest correlation of any two variables (correlation ≈ 0.39), suggesting that smoking is more common among older people.

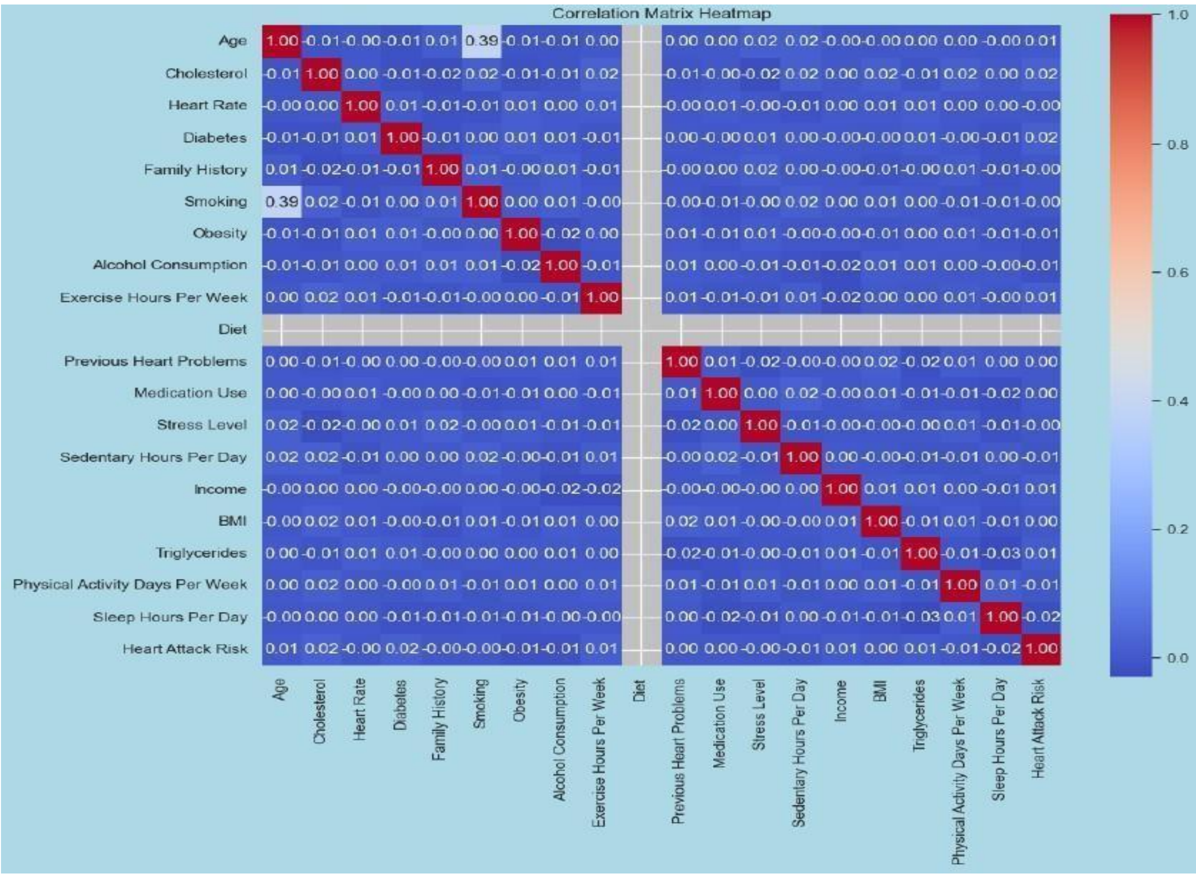


Figure 9: This figure is a correlation matrix heatmap

7. IDENTIFYING IMPORTANT COFACTORS

```
✅ Top 10 Selected Features:
Index(['Age', 'Cholesterol', 'Diabetes', 'Smoking', 'Obesity',
       'Alcohol Consumption', 'Exercise Hours Per Week', 'Income',
       'Triglycerides', 'Sleep Hours Per Day'],
      dtype='object')

📊 All Feature Rankings:
Age                                1
Triglycerides                      1
Income                             1
Alcohol Consumption                1
Obesity                            1
Exercise Hours Per Week            1
Diabetes                           1
Cholesterol                        1
Smoking                            1
Sleep Hours Per Day                1
Sedentary Hours Per Day            2
Physical Activity Days Per Week    3
Heart Rate                         4
Stress Level                       5
Medication Use                     6
Family History                     7
Previous Heart Problems            8
BMI                                9
dtype: int32
```

The process of feature selection was used to determine which variables were most pertinent to the model. Age, cholesterol, diabetes, smoking, obesity, alcohol consumption, exercise hours per week, income, triglycerides, and sleep hours per day were the top ten features chosen based on priorities. These characteristics were kept because they were the most relevant (rank = 1) and helped to increase interpretability, decrease complexity, and improve model accuracy

8. FITTING STATISTICAL MODELS FOR THE ANALYSIS

8.1 Logistic Regression

The classification algorithm known as logistic regression is applied when the dependent variable is binary, such as whether a person had a heart attack (1) or not (0).

Actions to take:

1. Bring in the necessary libraries.
2. Prepare the dataset by loading it.
3. Divide the Info
4. Fit the Model of Logistic Regression
5. Review the Model
6. Interpretations and Coefficients

Heart attack prediction was done using the logistic regression model. based on selected health and lifestyle factors. The following output was obtained:

```
✅ Accuracy: 0.6417569880205363

📊 Confusion Matrix:
 [[1125    0]
 [ 628    0]]

📄 Classification Report:
              precision    recall  f1-score   support

         0.0       0.64      1.00      0.78      1125
         1.0       0.00      0.00      0.00       628

    accuracy                           0.64      1753
   macro avg       0.32      0.50      0.39      1753
weighted avg       0.41      0.64      0.50      1753
```

The model achieved an overall accuracy of 64.17%, meaning that 64.17% of the total predictions were correct.

The confusion matrix showed the following results:

	Predicted: No Risk	Predicted: Risk
Actual: No Risk	1125	0
Actual: Risk	628	0

It was observed that the model successfully identified all instances of 'No Heart Attack Risk' (class 0) but failed to correctly identify any 'Heart Attack Risk' cases (class 1). The classification report further highlights this imbalance:

Class	Precision	Recall	F1-Score	Support
No Heart Attack Risk (0)	0.64	1.00	0.78	1125
Heart Attack Risk (1)	0.00	0.00	0.00	628

- Precision for class 0 was 64%, meaning that when the model predicted 'No Risk', it was correct 64% of the time.
- Recall for class 0 was 1.00, meaning that all actual 'No Risk' cases were identified.
- The model showed zero performance in detecting actual 'Heart Attack Risk' cases, with both precision and recall equal to 0 for class 1.

8.1.2. **ROC Curve and AUC Score**

The performance of a binary classifier is given by the ROC curve.At different threshold levels, it displays the True Positive Rate (TPR) against the False Positive Rate (FPR).

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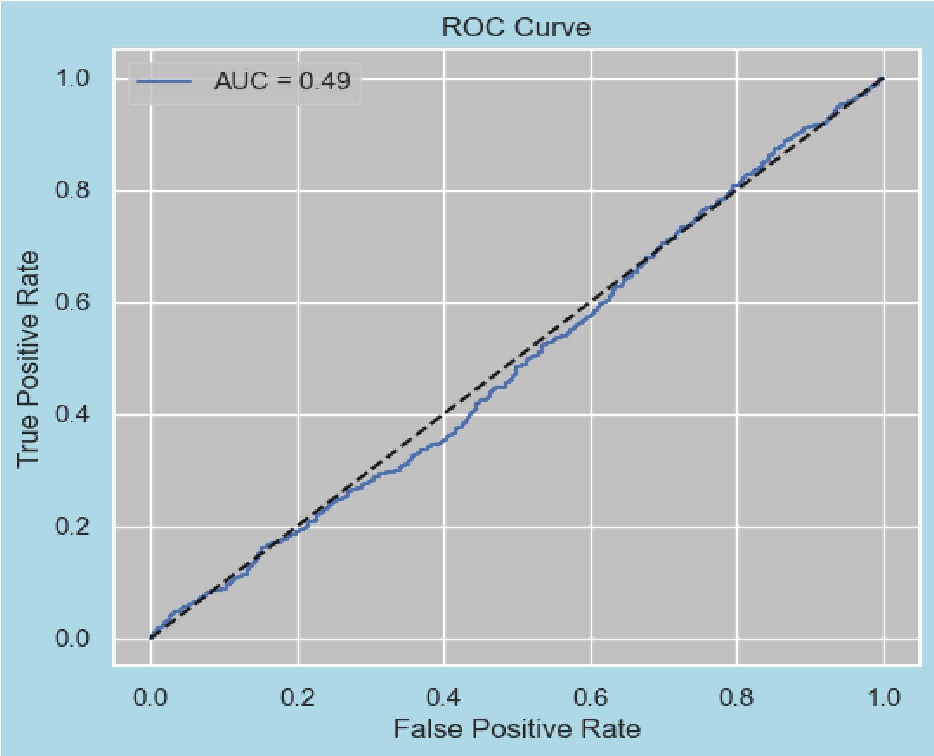

Fig 10: ROC curve showing logistic regression model performance

Since a random classifier has an AUC of 0.5, an AUC score of 0.49 means that the model performs worse than or about the same as random guessing.

Random performance is represented by the diagonal line (dashed line), which is very near to the ROC curve.

An AUC that is nearer 1.0 indicates that the model is good at differentiating between the two classes.

<u>In conclusion,</u>

Because of the dataset's class imbalance (more occurrences of "No Risk" than "Risk"), the logistic regression model was skewed toward predicting "No Heart Attack Risk." The model wasn't successful in

- An AUC score of 0.49 indicates that the model performs worse than or roughly equal to random guessing (since a random classifier has an AUC of 0.5).
- The ROC curve is very close to the diagonal line (dashed line), which represents random performance.
- A good model would have an AUC closer to 1.0, showing that it can successfully distinguish between the two classes.

Conclusion:

The logistic regression model was biased towards predicting 'No Heart Attack Risk', likely due to class imbalance in the dataset (more instances of 'No Risk' compared to 'Risk'). The positive cases of heart attacks was not classified by the model.

8.2 <u>Random Forest Classification</u>

A learning algorithm that creates multiple decision trees and then merges them is called random forest.

STEPS:

1. Import Required Libraries
2. Load and Prepare the Data
3. Fit the Random Forest Model
4. Fit the Random Forest Model

The Random Forest Classifier was used to predict heart attack risk, and the following results were obtained:

```
✅ Random Forest Accuracy: 0.638904734740445

📊 Confusion Matrix:
 [[1092   33]
 [ 600   28]]

📄 Classification Report:
              precision    recall  f1-score   support

         0.0       0.65      0.97      0.78      1125
         1.0       0.46      0.04      0.08       628

    accuracy                           0.64      1753
   macro avg       0.55      0.51      0.43      1753
weighted avg       0.58      0.64      0.53      1753
```

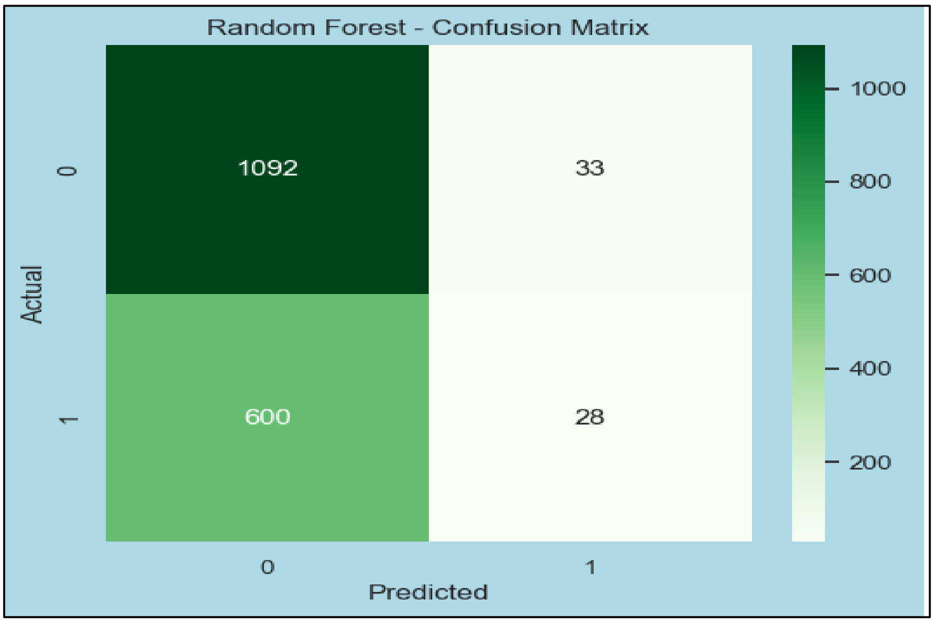

Fig 11: Confusion matrix for random forest

The Random Forest model achieved an overall accuracy of 63.89% on the test set.

The confusion matrix is as follows:

	Predicted: No Risk (0)	Predicted: At Risk (1)
Actual: No Risk (0)	1092	33
Actual: At Risk (1)	600	28

From this matrix, it is evident that:

- The model was able to correctly identify 1092 cases of patients not at risk (True Negatives).
- There were 33 false positives, where the model incorrectly classified patients as being at risk.
- The model failed to detect 600 true positive cases, classifying them as not at risk (False Negatives).
- Only 28 true positives were correctly identified as being at risk.

The detailed classification report is as follows:

Class	Precision	Recall	F1-Score	Support
0 (No Risk)	0.65	0.97	0.78	1125
1 (At Risk)	0.46	0.04	0.08	628
Accuracy			0.64	1753
Macro Avg	0.55	0.51	0.43	1753

Class	Precision	Recall	F1-Score	Support
Weighted Avg	0.58	0.64	0.53	1753

Interpretation:

- With a recall of 0.97 for Class 0, the model performs well in identifying patients who are not at risk.
- On the other hand, with a recall of just 0.04 for Class 1, the model does not perform well in identifying patients who are at risk of having a heart attack.

The model has trouble accurately classifying positive cases, as evidenced by the extremely low overall F1-score (0.08) for at-risk patients (Class 1).

Since the model tends to favor predicting the majority class (no risk), the class imbalance in the dataset is probably a contributing factor to this skewed performance

8.2.1 **ROC Curve and AUC Score**

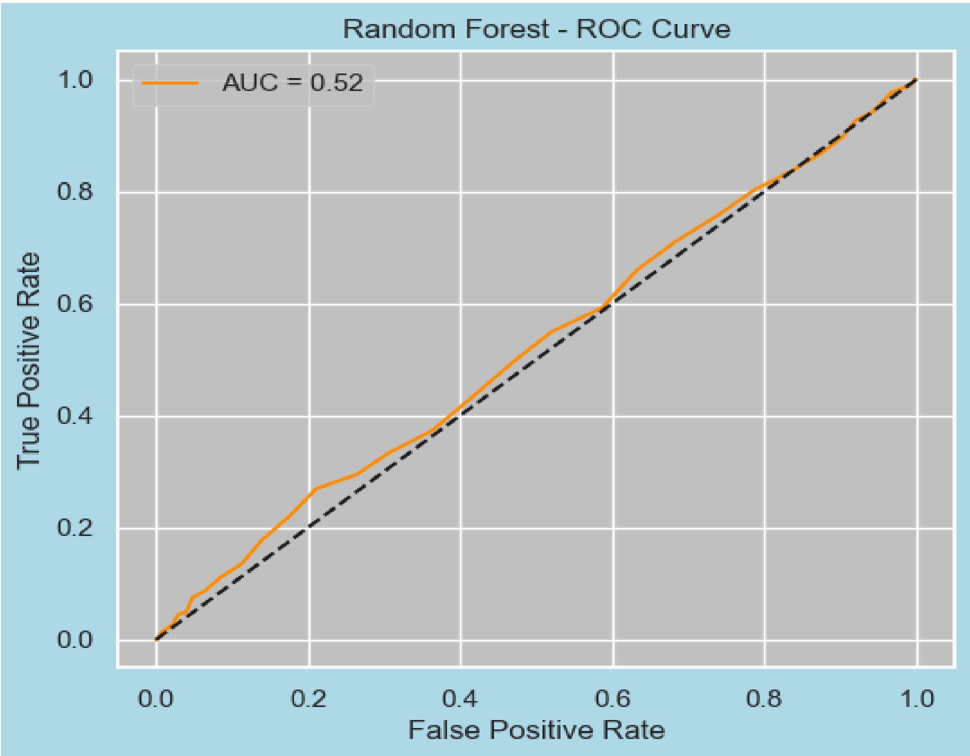

Fig 12: ROC curve for random forest model

The model is not successfully differentiating between the positive (at-risk) and negative (not at-risk) classes, as further evidenced by the ROC curve's close proximity to the diagonal line (represented by the dashed black line).

AUC should be nearer 1.0 in the best-case scenario, when the ROC curve rises sharply toward the top-left corner (high true positive rate, low false positive rate).

In conclusion:

Based on the provided dataset, the Random Forest model does not exhibit strong discriminative power in predicting the risk of a heart attack. The extremely low AUC indicates that the model cannot reliably distinguish patients at very low AUC suggests that the model is unable to confidently separate patients at risk from those not at risk.

9. COMPARISION OF THE MODEL PERFORMANCE

Evaluating the accuracy of two well-known classification algorithms in estimating the risk of a heart attack: Random Forest Classifier and Logistic Regression. The same dataset and assessment metrics were used to train and test both models.

Metric	Logistic Regression	Random Forest
Accuracy	~ 0.64	0.6389
Precision (Class 1 - Risk)	0.00	0.46
Recall (Class 1 - Risk)	0.00	0.04
F1-Score (Class 1 - Risk)	0.00	0.08
AUC (ROC Curve)	~ 0.52 (low)	0.52

- When identifying patients at risk of a heart attack, Random Forest outperformed Logistic Regression (Class 1).
- Class 1 precision increased to 0.46 (Random Forest) from 0.00 (Logistic Regression).
- Recall remains a challenge for both models, suggesting that they have trouble accurately identifying the majority of positive cases.
- Both models' accuracy was comparable (about 64%), but because of the class imbalance, accuracy is not the best metric in this case.
- Since the AUC for both models was low (~0.52), neither model is currently producing predictions with high confidence

10. **CONCLUSION**

Using a heart attack prediction dataset, a thorough analysis was conducted in this report to determine and comprehend the major factors influencing heart attack risk. Feature selection, correlation analysis, exploratory data analysis, and the use of a logistic regression model were all steps in the process. Heart attack risk was found to be significantly influenced by a number of factors, including age, income, sleep patterns, exercise hours, triglycerides, smoking, obesity, diabetes, cholesterol, and triglycerides. The logistic regression model obtained an accuracy of 64.17% in predicting the lack of heart attack risk, but was unable to correctly classify positive heart attack risk cases due to class imbalance in the dataset. This project illustrates that, while logistic regression gives a useful essential model, further effort is required to improve its predictive performance. Overall, the project offers valuable insights into the factors affecting heart attack risk and establishes a foundation for future research to create more accurate prediction models.

11. REFERENCES

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